FLEET TEAM DIGEST

737MAX-FTD-34-22009

Issue Title : Flight Management Computer (FMC) Reverts to 'Single FMC Operation' Due to Cross Comparison Failure

Airplane Model	ATA	Minor Model(s)
737MAX	3461-00	-10, -7, -8, -8200, -9, -BBJ
Other Model(s)	**ECCN**	
	9E991	

Originated Date	Last Revised Date	Created On
04/13/2023	03/27/2026	01/16/2026
Estimated Completion Date	**Next Update Date**	
01/31/2028	09/30/2026	

Revision Description

Updated the following section(s):

- Milestones

Description

Background

737 MAX operators (including the 737-10 Flight Test Program) with dual FMC installations using FMC OPS U14.1 software have reported the following: Delayed CDU keypress processing by the FMCs, Display of the SINGLE FMC OPERATION scratchpad message, and Diverging Captain and First Officer Navigation Displays (NDs). These occurrences are due to a change made in U14.1 in the way the FMC processes N1 Limit Mode changes from the Flight Control Computer (FCC) to address an issue in the U14.0 software where the FMC N1 Limit Mode could get stuck in ‘Cruise’ following a level-off during a climb. In this U14.0 behavior, the FMC would not process a change to ‘Climb’ N1 Limit Mode.

N1 Limit Modes are sent from the FCC to the FMC. The FCC commands the ‘Descent’ N1 Limit Mode at top of descent and the start of a step-descent and commands the ‘Cruise’ N1 Limit Mode at level-offs following the top of climb, step-climbs, and step-descents.

Technical Details

CDU keystrokes made at the same time as the FCC commands ‘Cruise’ or ‘Descent’ N1 Limit Modes may interrupt FMC synchronization and cause an event timeout in the secondary FMC. The FMC response to an event timeout is to initiate a Random Access Memory (RAM) crossload, since a successful RAM crossload will resolve any potential disagreements. A RAM crossload is where the primary FMC sends a copy of its RAM to the secondary FMC, in order to recover FMC synchronization. RAM crossloads are transparent to the flight crew, and are part of normal FMC behavior.

When the FMCs receive multiple CDU keypresses while the secondary FMC is waiting for synchronization, the secondary FMC’s

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keypress handling and synchronized event handling may compete, creating a situation where successful RAM crossloads are prevented. Depending on how this situation is resolved, and the timing of it, different behaviors may result.

Behaviors

The behavior that occurs depends on the timing of how the FMC synchronization and RAM crossloads fail.

1. FMC CDU keystroke processing is latent
   - An initial crossload will fail but a crossload 12-15 seconds later will resolve the issue with no other effects. The crew and system may experience slow, delayed CDU responses during this 12-15 second period of time, but will not experience any other flight deck effects.
   - The FMCS BITE will indicate a RAM crossload failure and FCC_N1_Limit_Mode Event Timeout(s).
2. SINGLE FMC OPERATION message
   - An initial crossload and another crossload will fail resulting in a SINGLE FMC OPERATION message.
   - If Autopilot (A/P) CMD-B is engaged, LNAV/VNAV engagement will be lost per requirements. The FMCS BITE will indicate RAM crossload failure(s), FCC_N1_Limit_Mode Event Timeout(s), and SINGLE FMC OPERATION.
3. Diverging Captain and First Officer Navigation Displays (NDs)
   - A cycle of four or more timeouts and crossloads will occur that may cause the secondary FMC to no longer respond to events and keystrokes. It has the possibility of self-recovery or latching.
   - A SINGLE FMC OPERATION message will not display as the crossloads are successful and the FMCs consider themselves to still be in dual operation. However, the secondary FMC's flight plan will not respond to edits and will not sequence waypoints, allowing the Captain and First Officer's flight plans to diverge. If Autopilot (A/P) CMD-B is engaged, the secondary FMC is the one coupled to the A/P and Flight Directors (F/D) and could result in unexpected turns as the secondary FMC tracks to its un-updated flight plan.
   - For this behavior, the FMCS BITE will indicate FCC_N1_Limit_Mode Event Timeout(s) and Key(s) Not Processed but will not indicate SINGLE FMC OPERATION.

Following difficulties resyncing resulting in behaviors 2 or 3, the secondary FMC may reset as it attempts to process data out of context. This results in a single FMC reset that recovers shortly and the primary FMC is not impacted.

Status

These conditions have been reproduced and their root cause has been determined. Boeing currently plans to incorporate the fix for this condition in FMC OPS U14.2. At this writing, the incorporation schedule is still not settled.

Interim Action

Avoiding CDU keystrokes during portions of flight where the FCC N1 Limit Mode changes to 'Cruise' or 'Descent' will help prevent these behaviors. Avoiding CDU keystrokes may not be operationally feasible in all cases. Some mitigating actions can be taken, depending on which behavior occurs:

1. FMC CDU keystroke processing is latent
   - If the crew notices CDU keystrokes are lagging or slow to process, this indicates that the FMCs are synchronizing. Pausing keystrokes when this behavior is observed will help prevent the situation from worsening.
2. SINGLE FMC OPERATION message
   - When the SINGLE FMC OPERATION message displays, follow the Quick Reference Handbook (QRH) task for FMC FAIL to restore dual FMC operation. Modification of the FMC Source Select Switch per the QRH task will force a RAM crossload

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and will resolve the situation.

3. Diverging Captain and First Officer Navigation Displays (NDs)
   - This situation may resolve itself as the FMCs attempt periodic resynchronization. If the crew notices a difference in the Captain and First Officer NDs, caution should be taken if using Autopilot B since the secondary FMC may not process flight plan edits or sequence waypoints.
   - The flight crew can choose to change the FMC Source Select Switch to BOTH ON L in order for the First Officer's ND to display correct data and for Autopilot B to be controlled by the primary FMC. It should be noted that doing so will stop the periodic resynchronization attempts between the FMCs. Changing the FMC Source Select Switch to BOTH ON R should not be done if this behavior occurs, since it may crossload incorrect flight plan data from the Right FMC to the Left FMC.

There are no maintenance tasks to perform since the current software is working as implemented. Following a suspected occurrence of this issue, FMC BITE data may be downloaded and sent to Boeing and GE Aviation for confirmation.

Final Action

Incorporate the U14.2 service bulletin on in-service aircraft.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

FMC OPS U14.2 - 737 MAX - Certification | No earlier than 2028 (Target)

FMC OPS U14.2 - 737 MAX - Production Incorporation | No earlier than 2028 (Target)

FMC OPS U14.2 - 737 MAX - Service Bulletin Release | No earlier than 2028 (Target)

Related Categories

- Flight Deck Effects and Maintenance Messages : Issue that occurs frequently and causes a flight deck effect or maintenance message

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